Problem Statement

Evaluate the following limit from the image "dailyintegral-limit-limit-1x1-2026-07-07.jpg":

$$\lim_{n\to\infty} \sqrt[n]{\operatorname{lcm}(1, 2, \ldots, n)}$$

Detailed Solution

This limit can be beautifully evaluated by expressing the least common multiple (LCM) in terms of its prime factorization and utilizing the Prime Number Theorem. Let's break this down step-by-step.

Step 1: Prime factorization of the LCM

Let $L_n = \operatorname{lcm}(1, 2, \ldots, n)$. We need to determine the prime factorization of L_n.

For any prime number $p \leq n$, the highest power of p that divides any integer up to n is p^k, where k is the largest integer such that $p^k \leq n$. Taking the base-p logarithm of both sides, we get $k \leq \log_p(n)$. Since k must be an integer, it is given by the floor function: $k = \lfloor \log_p(n) \rfloor$.

Therefore, the prime factorization of L_n is the product of $p^{\lfloor \log_p(n) \rfloor}$ over all prime numbers $p \leq n$:

$$L_n = \prod_{p \leq n} p^{\lfloor \log_p(n) \rfloor}$$

Step 2: Relate to Chebyshev's Second Function

To handle the n-th root in the limit, it is convenient to take the natural logarithm of L_n:

$$\ln(L_n) = \ln\left(\prod_{p \leq n} p^{\lfloor \log_p(n) \rfloor}\right)$$

Using the properties of logarithms, the logarithm of a product is the sum of the logarithms:

$$\ln(L_n) = \sum_{p \leq n} \lfloor \log_p(n) \rfloor \ln(p)$$

This summation is exactly the definition of Chebyshev's second function, widely used in analytic number theory and denoted as $\psi(n)$. Thus, we have the elegant relation:

$$\ln(L_n) = \psi(n)$$

Step 3: Apply the Prime Number Theorem

We are tasked with evaluating the limit:

$$L = \lim_{n\to\infty} \sqrt[n]{L_n} = \lim_{n\to\infty} (L_n)^{\frac{1}{n}}$$

Taking the natural logarithm of the entire limit gives:

$$\ln(L) = \ln\left(\lim_{n\to\infty} (L_n)^{\frac{1}{n}}\right) = \lim_{n\to\infty} \frac{\ln(L_n)}{n}$$

Substituting our result from Step 2 into the numerator:

$$\ln(L) = \lim_{n\to\infty} \frac{\psi(n)}{n}$$

A well-known, equivalent statement of the Prime Number Theorem (PNT) is that Chebyshev's second function is asymptotic to n as n approaches infinity. That is, $\psi(n) \sim n$. Consequently, the limit of their ratio is exactly 1:

$$\lim_{n\to\infty} \frac{\psi(n)}{n} = 1$$

So, we have found that $\ln(L) = 1$.

Step 4: Compute the final limit

Since the natural logarithm of our target limit is 1, we simply exponentiate to find the value of L:

$$L = e^1 = e$$

Final Answer:

$$\mathbf{e}$$